



UNIVERSITY OF MORATUWA
Faculty of Information Technology
B.Sc. (Hons) in Information Technology
B.Sc. (Hons) in Information Technology and Management
B.Sc. (Hons) in Artificial Intelligence
Semester 3 Examination

IN 2211 – OBJECT-ORIENTED ANALYSIS AND DESIGN

Time Allowed: 3 hours

January 2025

INSTRUCTIONS TO CANDIDATES

1. This paper contains 4 questions on 5 Pages, including this page.
2. The total marks obtainable for this examination is 100.
3. The marks assigned for each question are included in square brackets.
4. This examination accounts for 70% of the module assessment.
5. This is a closed-book examination.
6. Start answering each question using a new page.
7. Answer ALL questions.

ADDITIONAL MATERIAL

None.

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Question 01 [30 marks]

- a) Consider the below description regarding an e-commerce platform.

"ShopEasy" is an e-commerce platform that aims to provide customers with a seamless online shopping experience while supporting sellers in managing their products and orders. Customers can register, log in, and manage their profiles. They can browse product categories or search for specific items. Customers can view product details, including descriptions, reviews, ratings, and seller information. They can add items to their shopping cart, proceed to checkout, and make secure payments. Customers can track their order status, from confirmation to delivery. They should be able to view order history, request returns or refunds and provide feedback or reviews. Sellers can respond to this feedback or reviews. Sellers also can register and log in to create and manage their profiles. They can upload new products, update product details, and manage inventory. Sellers receive notifications for new orders and can update the order status (e.g., "Order Processed," "Shipped"). They can view sales reports and manage customer feedback. Administrators can monitor the platform's overall performance, including sales, user activity, and revenue. They can manage user accounts and handle disputes. When handling disputes, they review the evidence and issue resolutions (e.g., refunds, penalties). Administrators also can approve or remove sellers. They can generate reports and oversee promotional campaigns. They run platform-wide sales events such as flash sales and holiday discounts as promotional campaigns.

- I. Briefly explain the difference between the Include and Extend relationships.

[5 marks]

- II. Draw the UML Use Case Diagram to visualize the system.

[15 marks]

- b) Consider the below order process flow.

The customer selects items, adds them to the cart, proceeds to checkout, and completes payment. The system verifies the payment and confirms the order. The order enters the "Order Placed" state, and a notification is sent to the seller and customer. The seller reviews the order details and inventory. If the order is accepted, it transitions to the "Order Confirmed" state. If the order is rejected (e.g., item out of stock), it transitions to the "Order Cancelled" state, and the system initiates a refund. Once they enter order preparation, the seller updates the order status to the "Being Prepared" state. Once the order is prepared, the seller updates the status to "Ready for Pickup". The delivery personnel pick up the order from the seller. The system updates the order to the "Out for Delivery" state and sends a real-time tracking link to the customer. After the delivery personnel delivers the order to the customer, the order transitions to the "Delivered" state. The customer receives a confirmation notification and is prompted to provide feedback.

Continued...

If the customer is unsatisfied with the product, they can initiate a return or refund request. The order transitions to the "*Return Requested*" state and is reviewed by the seller or administrator. Upon approval, the order transitions to the "*Refund Processed*" state, and the payment is returned to the customer.

Draw a state diagram based on the provided description.

[10 marks]

Question 02 [25 marks]

Consider the following description of a Hospital Management System.

The Hospital Management System (HMS) must manage patients, doctors, appointments, and billing to facilitate the smooth functioning of a hospital. Patients can register with their personal details, including name, age, gender, address, and contact information. Each patient is assigned a unique ID and maintains a medical history. Each doctor has a unique ID, name, specialization, and availability schedule. Patients can book appointments with available doctors based on their schedule. Each appointment is assigned a unique ID, date, time, and status (e.g., "Scheduled," "Completed," "Cancelled"). Appointments may have additional notes or prescriptions from the doctor. After an appointment is completed, a bill is generated for the patient. The bill includes details such as consultation charges, hospital charges, and the total amount. Bills are linked to the patient's payment history. Administrators can manage doctor schedules and monitor appointments. They can also generate reports on hospital operations, such as patient visits, revenue, and doctor performance.

- a) Explain the difference between aggregation and composition by providing suitable examples.

[4 marks]

- b) Draw a UML Class Diagram for this Hospital Management System. Add attributes and methods when necessary. Visibility modifiers are not required.

[10 marks]

- c) Consider the below scenario of the hospital management system.

A patient named Kamal Perera (ID: P123) has scheduled an appointment with Dr. Ranga (ID: D456), a cardiologist, on January 15, 2025, at 10:00 AM. The appointment (ID: A789) is marked as "Scheduled" and includes a note: "Follow-up for chest pain." After the appointment, a bill (ID: B101) is generated with the following details:

Consultation Charges: Rs.3500

Hospital Charges: Rs. 1500

Total Amount: Rs. 5000

Draw the object diagram for the above scenario.

[6 marks]

Continued...

- d) Explain the importance of the package diagram when developing a hospital management system.

[5 marks]

Question 3 [25 marks]

Consider the following description.

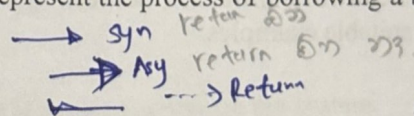
The University Library Management System is designed to streamline the process of borrowing and returning books, managing user accounts, and monitoring inventory. It involves interactions between students, librarians, and the system itself. Students can search and reserve books, borrow books, return books, and manage the account. A librarian can manage the book inventory, approve or deny book reservations, and handle fines. The system will generate reports and monitor the user activities within the system.

- a) Explain the significance of sequence diagrams and activity diagrams in system design. Highlight their unique roles and emphasize the key differences between the two diagrams.

[5 marks]

- b) When borrowing a book, the student has to search for a book using the catalog. Then the system will check the availability of the book. If the book is available, the student proceeds to borrow it by scanning their ID and the book barcode. The system updates the inventory, assigns the book to the student's account, and provides a borrowing receipt.

Draw the sequence diagram to represent the process of borrowing a book in the Library Management System.



[10 marks]

- c) When returning a book in the Library Management System, students can return the book at the library desk. The system verifies the book against the student's borrowing record. If the book is returned on time, the system updates the inventory and closes the borrowing record. If the book is returned late, the system calculates a fine and adds it to the student's account. The student has to pay the fine to the librarian. The librarian updates the system as fine paid and then the system generates the receipt. The student can obtain the receipt and complete the return process.

Draw the activity diagram to represent the process of returning a book in the Library Management System.

[10 marks]

Continued...

Question 04 [20 marks]

Consider the below description related to the Hotel Reservation System.

A Hotel Reservation System is designed to manage online bookings, room availability, customer interactions, and administrative tasks for a chain of hotels. Customers can search for and book rooms and manage reservations (e.g., view, modify, and cancel). They can also interact with a chatbot or contact customer support for inquiries or issues.

Hotel staff can update room availability, pricing, and special offers. Staff also can view and manage customer reservations and handle special requests. System administrators can manage users, monitor the system and configure promotions and campaigns.

- a) What are provided and required interfaces in component diagrams? [5 marks]
- b) It is designed to have seven (7) components in the Hotel Reservation System. **Customer Interface** handles customer interactions such as searching for rooms, managing reservations, and payments. **Hotel Management Component** is used by hotel staff to manage room availability, booking details, and reports. **Reservation Component** manages booking operations. **Payment Processing Component** integrates with payment gateways for processing transactions. The **Notification Component** manages all the notifications associated with the system. **Authentication Component** manages login, registration and role-based access control. **Database Component** stores data related to the system.

Draw the UML Component diagram for the Hotel Reservation System.

[10 marks]

- c) If a new feature, such as integration with third-party travel platforms, needs to be added, how would it impact the Component Diagram?

[5 marks]

End of the Paper.